

The Chinese Remainder Theorem for rings:

Consider the ring $(\mathbb{Z}, +, \cdot)$.

$$a \equiv b \pmod{3} \iff a - b \text{ is a multiple of } 3.$$

$$\text{Let } I = \{ \text{set of multiples of } 3 \} \\ = \{ 0, \pm 3, \pm 6, \dots \}.$$

$$\text{Hence } a \equiv b \pmod{3} \text{ means } a - b \in I.$$

Now let A, B be ideals of a ring R . Then

$$A+B = \{ a+b \mid a \in A, b \in B \} \text{ is an ideal. (verify!)}$$

let $m, n \in \mathbb{Z}$.

$$I = \{ \text{all multiples of } m \text{ in } \mathbb{Z} \}, \quad m \in I$$

$$J = \{ \text{ " " " " } n \text{ " " } \}, \quad n \in J$$

"Congruence modulo m " is the same as congruence modulo ideal I .

$$\text{If } (m, n) = 1, \text{ then } mu + nv = 1 \text{ for some } u, v \in \mathbb{Z}.$$

$$\Rightarrow m(ur) + n(vr) = r \quad (\text{by post multiply with } r \in \mathbb{Z}).$$

This means every integer $r =$ sum of a multiple of ' m ' and a multiple of ' n '.

$$= \text{sum of (an element of } I + \text{element of } J)$$

$$\text{Therefore } I+J = \mathbb{Z} \text{ (entire ring).}$$

This shows that the condition $(m, n) = 1$, says that $I+J = \mathbb{Z}$.

Note: When $(m, n) = 1$, then

$$I \cap J = IJ = \{ mu \mid \begin{matrix} m \in I \\ n \in J \end{matrix} \}$$

Note: Two Integers are congruent modulo 'n'
precisely when they are congruent modulo the
ideal $I \cap J$.

Theorem: (CRT for rings).

Let I and J be ideals of a ring R such that $I + J = R$.
Then for any $a, b \in R$, the system

$$\begin{cases} x \equiv a \pmod{I} \\ x \equiv b \pmod{J} \end{cases} \quad \text{say system } (*)$$

has solution. Any two solutions of the system are congruent
modulo $I \cap J$.

Proof. Since $I + J = R$ and $b - a \in R$, there exists
 $i \in I$ and $j \in J$ such that

$$b - a = i + j$$

$$\Rightarrow a + i = b - j$$

let $t = a + i$. Then $t - a = (a + i) - a = i \in I$

$$\Rightarrow t \equiv a \pmod{I}.$$

Since $a + i = b - j$, we have

$$t - b = (a + i) - b = (b - j) - b = -j \in J.$$

$$\Rightarrow t \equiv b \pmod{J}.$$

Therefore t is a solution of the system $(*)$.

If z is also a solution, then

$$z \equiv a \pmod{I}, \quad t \equiv a \pmod{I} \Rightarrow z \equiv t \pmod{I} \quad (\text{by transitive})$$

$$\text{Also } z \equiv b \pmod{J}, \quad t \equiv b \pmod{J} \Rightarrow z \equiv t \pmod{J}.$$

This means $z - t \in I$ and $z - t \in J$

$$\Rightarrow z - t \in I \cap J$$

$$\text{Hence } z \equiv t \pmod{I \cap J}.$$

Theorem: If I and J are ideals in a ring R and $I + J = R$, then there is an isomorphism of rings

$$R/(I \cap J) \cong R/I \times R/J$$

Proof. Define a map

$$f: R \rightarrow R/I \times R/J \text{ by}$$

$$f(r) = (r+I, r+J) \quad \forall r \in R.$$

$$f \text{ is a homomorphism: } f(r+s) = ((r+s)+I, (r+s)+J)$$

$$= (r+I, r+J) + (s+I, s+J)$$

$$= f(r) + f(s).$$

$$f(r) \cdot f(s) = (r+I, r+J) \cdot (s+I, s+J) = (rs+I, rs+J) = f(rs).$$

$$f \text{ is surjective: Take } (a+I, b+J) \in R/I \times R/J.$$

By CRT, $\exists t \in R \ni t$ is a solution of the system

$$x \equiv a \pmod{I},$$

$$x \equiv b \pmod{J}.$$

$$\text{But } t \equiv a \pmod{I} \Rightarrow t-a \in I \text{ i.e. } t+I = a+I.$$

$$\text{Similarly, } t \equiv b \pmod{J} \Rightarrow \dots \dots \dots t+J = b+J.$$

$$\text{Thus } f(t) = (t+I, t+J) = (a+I, b+J).$$

Thus by 1st Iso. Theorem,

$$R/\ker f \cong R/I \times R/J.$$

$$\text{where } \ker f = \{ r \in R \mid f(r) = 0_{R/I \times R/J} \}.$$

$$= \{ r \in R \mid (r+I, r+J) = (0_R+I, 0_R+J) \}$$

$$= \{ r \in R \mid r+I = 0+I, r+J = 0+J \}$$

$$= \{ r \in R \mid r \in I, r \in J \}$$

$$= \{ r \in R \mid r \in I \cap J \} = I \cap J.$$

Cor: If $(m, n) = 1$, then

$$\mathbb{Z}_{mn} \cong \mathbb{Z}_m \times \mathbb{Z}_n \quad (\text{isomorphic rings}).$$

Pf. let $(\mathbb{Z}, +, \cdot)$ be considered.

(m) : ideal consisting of all multiples of m .

(n) : " " " " " "

$$(m) + (n) = \mathbb{Z}.$$

$$\text{Also } (m) \cap (n) = (mn)$$

Therefore the quotient rings are respectively

$$\mathbb{Z}/(mn), \mathbb{Z}/(m), \text{ and } \mathbb{Z}/(n) \text{ are}$$

$$\mathbb{Z}_{mn}, \mathbb{Z}_m \text{ and } \mathbb{Z}_n.$$

$$\left| \begin{array}{l} \text{Here: } R = \mathbb{Z} \\ I = (m) \\ J = (n) \end{array} \right.$$

$$\mathbb{Z}_{mn} = \mathbb{Z}/(mn) = \mathbb{Z}/(m) \cap (n)$$

$$\cong \mathbb{Z}/(m) \times \mathbb{Z}/(n)$$

$$= \mathbb{Z}_m \times \mathbb{Z}_n.$$

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